

Lesson 13

Topic: Introduce time-based calculations.

Prerequisites: Download Chocolate Sales.csv file and disable Auto DateTime options by Options in Current File and create new DATE table by using Calendarauto and Format functions.

link:

C:\Users\MaabLLC\Desktop\maab_files\PowerBI_Homeworks\Chocolate Sales.csv

Basic Level (1–5)

- 1) Total Sales Amount (All-Time)

Load the file

File path:

C:\Users\MaabLLC\Desktop\maab_files\PowerBI_Homeworks\Chocolate Sales.csv

Open Power BI → Home → Import Data → Text/CSV → Select the file.

Disable Auto Date/Time

File → Options and settings → Options → Current File → Data Load → Uncheck Auto Date/Time.

Create a DATE Table

Go to Modeling tab → New Table:

DAX

```
Date = ADDCOLUMNS(
    CALENDARAUTO(),
    "Year", YEAR([Date]),
    "Month Number", MONTH([Date]),
    "Month Name", FORMAT([Date], "MMMM"),
    "Quarter", "Q" & FORMAT([Date], "Q"),
    "Weekday", FORMAT([Date], "dddd"),
    "Weekday Num", WEEKDAY([Date], 2)
)
```

Create relationship

Link Date[Date] to your fact table's Sales[Order Date] column.

Basic Level (1–5) – DAX Measures

1. Total Sales Amount (All-Time)

DAX

Total Sales = SUM('Chocolate Sales'[Sales Amount])

2. Total Sales for the Current Year

DAX

```
Total Sales YTD = CALCULATE(  
    [Total Sales],  
    YEAR('Date'[Date]) = YEAR(TODAY())  
)
```

Or better:

DAX

Total Sales YTD = TOTALYTD([Total Sales], 'Date'[Date])

3. Total Sales for Last Year

DAX

```
Total Sales Last Year = CALCULATE(  
    [Total Sales],  
    SAMEPERIODLASTYEAR('Date'[Date])  
)
```

4. Total Sales This Month

DAX

```
Total Sales This Month = CALCULATE(  
    [Total Sales],  
    MONTH('Date'[Date]) = MONTH(TODAY()) &&  
    YEAR('Date'[Date]) = YEAR(TODAY())  
)
```

5. Total Sales This Quarter

DAX

Total Sales This Quarter = TOTALQTD([Total Sales], 'Date'[Date])

- 2) Create a measure to calculate total sales for the current year.

Step-by-Step Guide

1. Make sure you have:

- A proper **Date table** (created with CALENDARAUTO() or similar)
- A **relationship** between your Date table and your sales table

2. Create the Total Sales measure:

Go to **Modeling** → **New Measure** and enter:

DAX

Total Sales =

SUM('Chocolate Sales'[SalesAmount])

This calculates **all-time** sales.

3. Now create the measure for Current Year:

DAX

Total Sales CY =

```
CALCULATE(  
    [Total Sales],  
    YEAR('Date'[Date]) = YEAR(TODAY())  
)
```

This filters the Date table to only include dates where the year matches the **current year**.

Or Better: Use DAX's DATESYTD function

(Recommended for time intelligence visuals)

DAX

Total Sales CY =

```
CALCULATE(  
    [Total Sales],  
    DATESYTD('Date'[Date])  
)
```

This version automatically handles the **start of the year to today** using calendar logic.

Example in Power BI:

1. Add a **Card visual**
2. Drag Total Sales CY into it
3. It will show total sales from **Jan 1st to today** (current year only)

- 3) Create a measure to calculate total sales for the last year.

Step-by-Step: Total Sales for Last Year

1. First, make sure you already have:

- A **Date table** using CALENDARAUTO()
- A base measure called **Total Sales** like this:

DAX

Total Sales =

SUM('Chocolate Sales'[SalesAmount])

2. Now create the Last Year Sales measure

Go to **Modeling** → **New Measure**, and paste this DAX:

DAX

Total Sales LY =

```
CALCULATE(  
    [Total Sales],  
    SAMEPERIODLASTYEAR('Date'[Date])  
)
```

How it works:

- CALCULATE(...): changes the context of calculation
- SAMEPERIODLASTYEAR(...): shifts the current filter context **exactly one year back**
- So, if you filter for 2025, this measure returns **sales from 2024**

Add to a Visual:

Use this in a **card** or **matrix** to compare with Total Sales CY (Current Year)

Optional: Show Year-over-Year Comparison

DAX

Sales YoY Change % =

```
DIVIDE(  
    [Total Sales CY] - [Total Sales LY],  
    [Total Sales LY],  
    0  
)
```

This gives you the % **growth or decline** from last year to this year.

Let me know if you want the same logic for:

- **Month-to-date vs last year**
- **Quarter-to-date vs last year**

- 4) Create a measure to calculate total sales for the current month.

Step-by-Step: Total Sales for the Current Month

1. Ensure you have:

A proper Date table created with CALENDARAUTO()

A relationship between Date[Date] and your sales table's date column

A base measure called:

DAX

Total Sales =

```
SUM('Chocolate Sales'[SalesAmount])
```

2. Create the Current Month Sales measure

Option 1: Using MONTH() and YEAR() functions

DAX

Total Sales CM =

```
CALCULATE(  
    [Total Sales],  
    MONTH('Date'[Date]) = MONTH(TODAY()) &&  
    YEAR('Date'[Date]) = YEAR(TODAY())
```

)

This measure filters the data to the current month and year.

Option 2 (Recommended): Using DATESMTD (Date Month-To-Date)

DAX

Total Sales CM =

CALCULATE(

[Total Sales],

DATESMTD('Date'[Date])

)

DATESMTD() gives you the range from the start of the current month to today.

Use in Report

Add a Card visual and insert this measure to show current month's total sales.

Can be used with comparison measures or trend charts too.

- 5) Create a measure to calculate total sales for the current quarter.

Step-by-Step: Total Sales for the Current Quarter

1. Make sure you already have:

- A **Date table** created with CALENDARAUTO()
- A base measure called Total Sales:

DAX

Total Sales =

SUM('Chocolate Sales'[SalesAmount])

2. Create the Current Quarter Sales measure

Option 1: Using DATESQTD (Recommended)

DAX

Total Sales CQ =

CALCULATE(

```
[Total Sales],  
    DATESQTD('Date'[Date])  
)
```

DATESQTD() = Date range from the **start of the current quarter to today**
Automatically handles correct quarter logic.

Option 2: Manual Logic (if not using DATESQTD)

DAX

Total Sales CQ =

```
CALCULATE(  
    [Total Sales],  
    QUARTER('Date'[Date]) = QUARTER(TODAY()) &&  
    YEAR('Date'[Date]) = YEAR(TODAY())  
)
```

This manually filters your data to dates within the same quarter as today.

Visual Example

- Use a **Card visual** to show Total Sales CQ
- Or use a **line/bar chart** to compare quarters over time

Optional: Quarter-over-Quarter Comparison

DAX

Total Sales LQ =

```
CALCULATE(  
    [Total Sales],  
    PARALLELPERIOD('Date'[Date], -1, QUARTER)  
)
```

Intermediate (6–10)

- 6) Sales Growth % Compared to Last Year (%YoY)
Step-by-Step

Step 1: You must already have these measures:

DAX

Total Sales =

SUM('Chocolate Sales'[SalesAmount])

Total Sales LY =

CALCULATE(

[Total Sales],

SAMEPERIODLASTYEAR('Date'[Date])

)

Step 2: Now Create the Sales Growth % YoY Measure

DAX

Sales Growth % YoY =

DIVIDE(

[Total Sales] - [Total Sales LY],

[Total Sales LY],

0

)

Explanation:

[Total Sales] = current period sales

[Total Sales LY] = same period last year's sales

DIVIDE(..., ..., 0) = handles division safely and returns 0 if denominator is blank

Output:

This measure shows:

Positive % → growth

Negative % → decline

0% → no change from last year

Use in Report:

Place this in a Card or Matrix visual

Add slicer for Year, Product, Region, etc.

- 7) Create a measure to calculate to retrieve sales from the last month.

Measure: Total Sales Last Month

Step 1: Make sure you already have:

- A proper Date table (created with CALENDARAUTO() or manually)
- A relationship between Date[Date] and the date column in your sales table
- A base sales measure:

DAX

Total Sales =

SUM('Chocolate Sales'[SalesAmount])

Step 2: Create the Last Month Sales measure

Recommended Method: Use PARALLELPERIOD

DAX

Total Sales LM =

CALCULATE(

[Total Sales],

PARALLELPERIOD('Date'[Date], -1, MONTH)

)

How it works:

- PARALLELPERIOD(..., -1, MONTH) shifts the current filter context **1 month back**
- This works with any visual that filters by date (like month or year slicers)

Alternate Method (Manual Date Logic):

DAX

Copy code

Total Sales LM =

CALCULATE(

[Total Sales],

MONTH('Date'[Date]) = MONTH(TODAY()) - 1 &&

YEAR('Date'[Date]) = YEAR(TODAY())

)

Not recommended for year boundaries (e.g., from January to December) because it won't handle the year change properly.

Optional: Add This to a Report

- Use a **Card visual** or **Matrix** to show Last Month's sales
- Use with Current Month sales to compare

- 8) Create a measure to calculate a running total of sales using

Running Total (Cumulative Sales)

A running total accumulates sales over time — for example, sales from January to June will add up each month:

makefile

Jan: 100

Feb: 100 + Feb sales

Mar: Jan + Feb + Mar sales

...

Step 1: Make sure you already have:

A proper Date table

A relationship between your Date table and the sales table

A base measure like this:

DAX

Total Sales =

SUM('Chocolate Sales'[SalesAmount])

Step 2: Create the Running Total (Cumulative Sales) measure

DAX

Running Total Sales =

CALCULATE(

[Total Sales],

FILTER(

ALLSELECTED('Date'[Date]),

```
'Date'[Date] <= MAX('Date'[Date])
)
)
```

Explanation:

ALLSELECTED('Date'[Date]): keeps your visual or slicer filters active

'Date'[Date] <= MAX('Date'[Date]): includes all dates up to the current row date

CALCULATE(...): re-calculates your sales using this custom filter

Use Case:

In a Line Chart or Table by month or date

You will see the values increase over time

- 9) Create a measure to compute sales for the last 3 months.

Measure: Total Sales for the Last 3 Months

This calculates total sales for the **3 full months before the current month** — not including the current month.

Step 1: Prerequisites

Make sure you already have:

- A proper Date table
- A relationship between Date[Date] and your sales table
- A base measure:

DAX

Total Sales =

SUM('Chocolate Sales'[SalesAmount])

◆ Step 2: Create the "Sales Last 3 Months" Measure

DAX

Total Sales Last 3 Months =

CALCULATE(

```

[Total Sales],
DATESINPERIOD(
    'Date'[Date],
    EOMONTH(TODAY(), -1), -- End of last month
    -3, -- Going back 3 months
    MONTH
)
)

```

Explanation:

- EOMONTH(TODAY(), -1) → Gets the **end of the previous month**
- -3 → Goes back **3 months**
- DATESINPERIOD(...) → Returns a date range from the past 3 months
- CALCULATE(...) → Recalculates sales within that 3-month period

Example:

If today is **June 2025**, this measure will return sales from:

March 1, 2025 → May 31, 2025

(it skips June because it includes **only** the 3 *previous* months)

How to Use:

- Show it in a **Card** visual
 - Use it in a **line or bar chart** with comparison to current sales
- 10) Create a measure to identify the month with the highest sales in the previous 12 months.

Step 1: Prerequisites

You should already have:

- A proper Date table
- A base measure:

DAX

Total Sales =

SUM('Chocolate Sales'[SalesAmount])

Step 2: Create the measure for Highest Sales Month (Name)

DAX

Best Sales Month Last 12M =

VAR Last12Months =

 DATESINPERIOD('Date'[Date], TODAY(), -12, MONTH)

VAR SalesByMonth =

 ADDCOLUMNS(

 SUMMARIZE('Date', 'Date'[Year], 'Date'[Month Number], 'Date'[Month Name]),

 "Sales", CALCULATE([Total Sales], Last12Months)

)

VAR MaxSales =

 MAXX(SalesByMonth, [Sales])

VAR BestMonthRow =

 FILTER(SalesByMonth, [Sales] = MaxSales)

RETURN

MAXX(

 ADDCOLUMNS(

 BestMonthRow,

 "MonthLabel", 'Date'[Month Name] & " " & 'Date'[Year]

),

 [MonthLabel]

)

What it does:

1. **Last12Months** → Creates a date range for the past 12 months
2. **SalesByMonth** → Groups sales by month and year
3. **MaxSales** → Finds the highest sales amount
4. **BestMonthRow** → Filters for the row with that max sales
5. **Return** → Combines month name + year (e.g., "March 2025")

Example Output:

"**November 2024**" ← means that's the best month in terms of total sales in the last 12 months.

Tip:

- Use in a **card visual** to display the best month
- Or use together with Total Sales in Best Month if needed

Advanced (11–15)

- 11) Create a measure to compare Q1 Sales of Each Year. Use time intelligence functions with filters to isolate and compare Q1 across years.
Step-by-Step Approach

We'll use DAX Time Intelligence + filters to isolate Q1 and enable year-over-year comparison.

Base Sales Measure

Make sure you already have:

DAX

Total Sales =

SUM('Chocolate Sales'[SalesAmount])

Final DAX Measure: Q1 Sales (Per Year)

Q1 Sales =

CALCULATE(

[Total Sales],

FILTER(

'Date',

MONTH('Date'[Date]) >= 1 && MONTH('Date'[Date]) <= 3

)

)

Explanation:

CALCULATE(...): modifies the context to filter only Q1 months

`FILTER('Date', ...)`: limits the data to January, February, and March of each year

Works great in a matrix or bar chart with 'Date'[Year] on the axis

Optional: Compare Q1 Sales YoY (% Growth)

DAX

Q1 Sales LY =

```
CALCULATE(  
    [Q1 Sales],  
    SAMEPERIODLASTYEAR('Date'[Date])  
)
```

Q1 Sales YoY % =

```
DIVIDE(  
    [Q1 Sales] - [Q1 Sales LY],  
    [Q1 Sales LY],  
    0  
)
```

How to Visualize in Power BI:

Use a bar chart or matrix:

Axis: 'Date'[Year]

Value: Q1 Sales

Optional: Add Q1 Sales YoY % as a second value

- 12) Create a measure to Show YoY Difference Only for December.

1. Base Sales Measure

Make sure this measure exists:

DAX

Total Sales =

```
SUM('Chocolate Sales'[SalesAmount])
```

2. December Sales (Current Year)

DAX

December Sales =

```
CALCULATE(  
    [Total Sales],  
    MONTH('Date'[Date]) = 12  
)
```

3. December Sales (Last Year)

DAX

December Sales LY =

```
CALCULATE(  
    [Total Sales],  
    MONTH('Date'[Date]) = 12,  
    SAMEPERIODLASTYEAR('Date'[Date])  
)
```

4. Final Measure: YoY Difference Only for December

DAX

December YoY Difference =

```
IF(  
    MONTH(MAX('Date'[Date])) = 12,  
    [December Sales] - [December Sales LY],  
    BLANK()  
)
```

Explanation:

MONTH(MAX('Date'[Date])) = 12 → Ensures calculation runs only for December rows

Returns difference between this December and last December

BLANK() hides values for all other months

Optional: YoY % Difference for December

DAX

December YoY % =


```

IF(
    MONTH(MAX('Date'[Date])) = 12,
    DIVIDE([December Sales] - [December Sales LY], [December Sales LY], 0),
    BLANK()
)

```

Format this as percentage.

Use It In:

A Matrix visual with 'Date'[Year] and 'Date'[Month Name]

December rows will show YoY Difference

Other months will be empty (BLANK)

- 13) Create a measure that sums the last 12 months using.

Create a DAX measure that calculates the **total sales over the last 12 months** from today (a rolling 12-month total).

Step 1: Prerequisites

You must already have:

- A proper Date table
- A relationship between 'Date'[Date] and your sales table
- A base sales measure:

DAX

Total Sales =

SUM('Chocolate Sales'[SalesAmount])

Step 2: Final DAX Measure — Last 12 Months Sales

DAX

Total Sales Last 12 Months =

CALCULATE(

[Total Sales],

DATESINPERIOD(

```

    'Date'[Date],
    MAX('Date'[Date]), -- Anchor date (e.g., from chart)
    -12, -- Look back 12 months
    MONTH
)
)

```

Explanation:

- CALCULATE(...): changes the context of your calculation
- DATESINPERIOD(...): creates a 12-month window **ending on the current date**
 - MAX('Date'[Date]) uses the latest date in the visual context
 - -12, MONTH goes back exactly 12 months

Example:

If you're viewing data for **June 2025**, this measure sums sales from:

July 2024 → June 2025

How to Use:

- Place in a **line chart** with 'Date'[Month] on the axis
- Shows how 12-month total changes over time
- Can also be used in **KPIs, cards, or matrix visuals**

- 14) Create a measure to identify sales Difference Between Current Quarter and Previous Quarter

Sales in current quarter – Sales in previous quarter

Step 1: Base Measure

Ensure you have a base measure:

DAX

Total Sales =

SUM('Chocolate Sales'[SalesAmount])

Step 2: Measure – Sales in Current Quarter

DAX

Current Quarter Sales =

```
CALCULATE(  
    [Total Sales],  
    DATESQTD('Date'[Date])  
)
```

Step 3: Measure – Sales in Previous Quarter

DAX

Previous Quarter Sales =

```
CALCULATE(  
    [Total Sales],  
    DATESQTD(  
        DATEADD('Date'[Date], -1, QUARTER)  
    )  
)
```

Step 4: Final Measure – Quarter-over-Quarter Sales Difference

DAX

QoQ Sales Difference =

[Current Quarter Sales] - [Previous Quarter Sales]

Explanation

- DATESQTD(...) → Gets the **quarter-to-date** range
- DATEADD(..., -1, QUARTER) → Shifts the date context back 1 quarter
- Final measure subtracts **previous quarter sales** from **current quarter**

Optional: Percent Difference (Growth %)

DAX

QoQ Sales Growth % =

```
DIVIDE(  
    [Current Quarter Sales] - [Previous Quarter Sales],  
    [Previous Quarter Sales],  
    0
```

)

 Format this as a **percentage** in the Modeling tab.

Where to Use:

- **Matrix** with 'Date'[Year] and 'Date'[Quarter]
- **Line chart** showing trends over quarters
- **Cards** for quick KPI view

- 15) Create a measure to highlight Months Where Sales Exceeded Previous Year by 10%+
 - Compare monthly totals and return a flag when current > 110% of last year.

Compare each month's sales to the same month last year, and return a value (e.g., "Above 10% Increase" or 1) **only if** this year's sales > **110%** of last year's.

Step 1: Base Measures

Ensure these exist:

DAX

Total Sales =

SUM('Chocolate Sales'[SalesAmount])

DAX

Sales LY =

CALCULATE(

[Total Sales],

SAMEPERIODLASTYEAR('Date'[Date])

)

Step 2: Final Measure — Highlight Months with >10% YoY Increase

DAX

Highlight 10%+ Growth =

IF(

NOT ISBLANK([Sales LY]) &&

DIVIDE([Total Sales], [Sales LY], 0) > 1.1,

```
"▲ Above 10%",  
BLANK()  
)
```

Explanation:

- SAMEPERIODLASTYEAR(...) gives last year's same month
- DIVIDE([Total Sales], [Sales LY], 0) calculates the growth ratio
- If ratio > **1.1** (i.e., 110%), it flags the row with "▲ Above 10%"
- Otherwise, returns blank

Alternative (Numeric Flag Version):

DAX

Highlight 10%+ Growth (Flag) =

```
IF(  
    NOT ISBLANK([Sales LY]) &&  
    DIVIDE([Total Sales], [Sales LY], 0) > 1.1,  
    1,  
    0  
)
```

You can use this in **conditional formatting** or filters.

How to Use:

- Put in a **matrix visual** with 'Date'[Month] and 'Date'[Year]
- Add Total Sales, Sales LY, and Highlight 10%+ Growth
- Or use conditional formatting based on this measure